

AuK

An Open-Source Foundational Model for Speech Generation and Editing



1 Speech Generation



Instruct TTS

Read the following text as a **warm, elderly storyteller, with a slightly raspy voice and a gentle, unhurried pace**:
"Long ago, beneath the southern lights, a little penguin learned that courage begins with one small step."

"Long ago, beneath the southern lights, a little..."



Zero-shot TTS

Ref Audio: [Audio waveform]
Say the following with the same voice: "Welcome to the future of sound, where every idea can be heard, shaped, and shared..."

"Welcome to the future of sound, where every..."



2 Acoustic Editing



Speed

Faster [Audio waveform]
Change the speed faster.

Slower [Audio waveform]
Change the speed slower.

Volume

Louder [Audio waveform]
Make the audio louder.

Softer [Audio waveform]
Make the audio softer.

Pitch

Higher [Audio waveform]
Raise the pitch higher.

Lower [Audio waveform]
Lower the pitch.

3 Paralinguistic Editing



Emotion

Happy [Audio waveform] Sad [Audio waveform]
Angry [Audio waveform] Calm [Audio waveform]

De-accent

Speech with accent [Audio waveform]
Remove the speaker's regional accent.
Standard Mandarin [Audio waveform]

Nonverbal

Add Nonverbal Event [Audio waveform]
[sigh] [cry]
Remove Nonverbal Event [Audio waveform]
[cough] [laugh]

Timbre

Original Timbre [Audio waveform]
Change the vocal timbre to fit this description: "The speaker has..."
Edited Timbre [Audio waveform]

Whisper

Normal [Audio waveform]
normal2whisper [Audio waveform]
Whisper [Audio waveform]
whisper2normal [Audio waveform]

4 Content Editing



Speech Content Editing

Input [Audio waveform]
"The old intro starts after the greeting and plays softly."
Edited [Audio waveform]
"The **old** intro starts **after** **before** the **warm** greeting and plays softly."



Music Lyric Editing

Input [Audio waveform]
"We walk alone beneath the city lights tonight."
Edited [Audio waveform]
"We **dance together** beneath the **neon** lights tonight."



5 Enhancement and Separation



Speech Separation

Mixed Speakers [Audio waveform]
Extract the speaker who says: "xxx".
Target Speaker [Audio waveform]

Music Separation

Mixed Music [Audio waveform]
Extract the vocal track.
Vocal [Audio waveform]

Speech Enhancement

Noisy Speech [Audio waveform]
Enhance the noisy speech.
Clean Speech [Audio waveform]